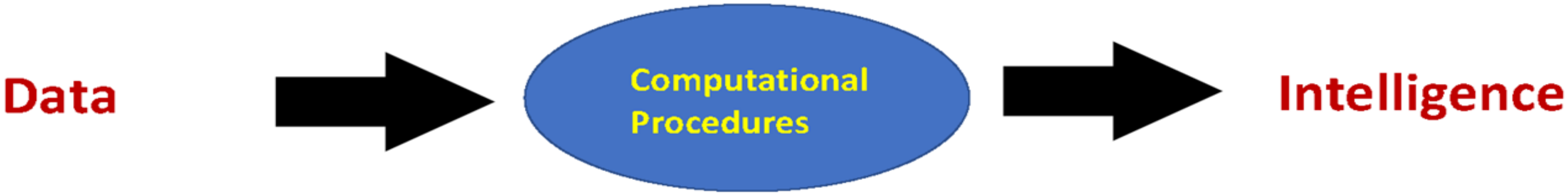
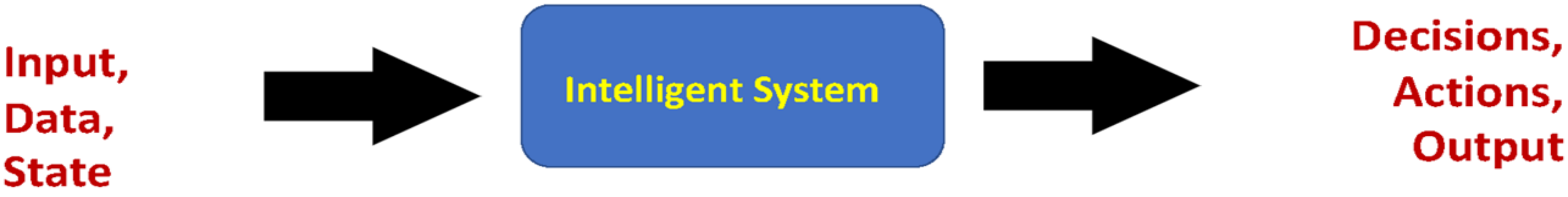
Learning to Practice

What is Modern “AI” or “ML”?



Data Driven Intelligence



Emerging Space of AI

Domains, Applications and Problems

- Flood Prediction & Mitigation
- Safety on Roads
- Technology for Specially, Abled
- Fraud Detection in Banks
- Low Cost Medical Screening
- Crowd Manag. (Public Safety)
- Disease Prediction & Prevention
- Education/Skilling in Indian Language
- Search Engine for Legal Judgment
- Air Quality Measurement

Algorithms and Areas

- Computer Vision
- Speech Recognition
- Language Processing
- Robotics
- Classification
- Clustering
- Regression
- Sequential Data Analysis
- Other Machine Learning

Data and Technology

- Data Annotation
- Data Platforms
- General (people) Internet Including Social Media
- CPS (Sensor Networks)
- Remote Sensing
- Internet of Things
- Mobile Devices
- E – Governance & Workflow Systems

The Powerful Hammer (we have)

- Data:
  - Plenty of “labelled” Data
- Architecture
  - A neural network architecture (eg. RNN, ResNet; Encoder-Decoder, Transformer)
- Loss Function
  - A goal for training
- A Training algorithm
  - Backpropagation (only one enough?) with many fine tricks
- Hardware-Software Infra
  - GPUs; Cloud; Cluster; TF; PyTorch
- A Training Strategy
  - Pre-Train; Self-Supervised Pre-Train

Is this (as) simple?

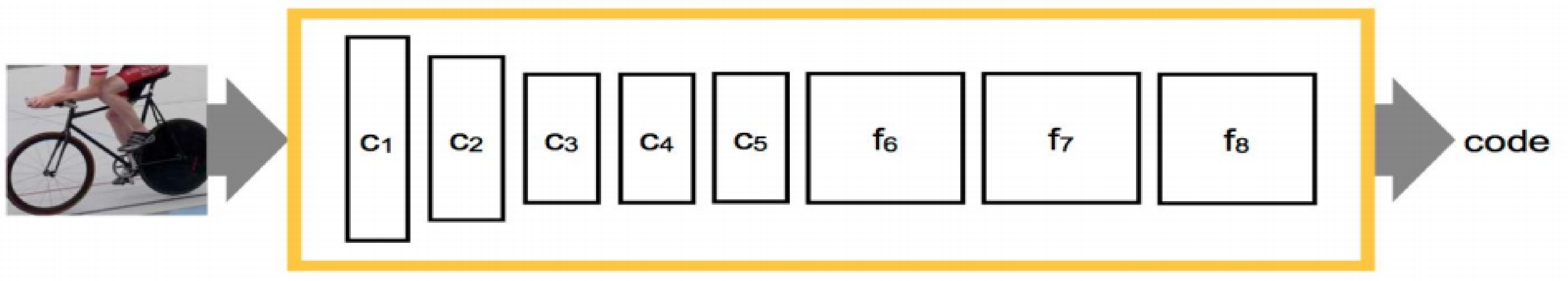
- We use “google” data; ”amazon” compute and “facebook” code/libraries
  - Is that so? What is for us to do then?
- We can just collect data, tag/annotate, and ask Neural network to learn.
  - Is that so simple? No role for us? Why are problems not getting solved?
- What do we tag/annotate? It is a simple human task, and India is not shot of human resources.
  - Is that so simple?
- All standard algorithms and architectures (BP, CNN, Tranformers etc.) are all available in public/popular libraries
  - Nothing further to do?

Examples of Practical Issues

- Amount of Data available for Learning
  - Very small amount is available in many situations
  - Few shot learning; Zero shot learning, One shot learning
- Limited Supervision
  - Amount of supervision (small annotated and more unannotated)
  - Cost of annotation (how do we minimize the cost; human in the middle)
  - Type of annotation (Eg. weak annotation and noisy annotation)
- Newer Situations
  - Training was done in a different dataset
  - Eg. Domain adaptation, Transfer learning

Learning Under Practical Constraints

Recap: Learned Representations



New Settings

- Extend to more classes
  - Extend from 1000 classes (say people) to another new 100
- Extend to new tasks
  - Extend from object classification to scene classification
- Extend to new data sets
  - Extend from imageNet to PASCAL (SLR to webcams)
- When we have a lesser amount of data.

Learned Representations

CNN Features can be used for wider applications:

- Train the CNN (deep network) on a very large database such as imageNet.

- Re-use CNN to solve smaller problems
  – Remove the last layer (classification layer)
  – Output is the code/feature representation

Transfer learning

- Same domain, different task
- Pre-trained Image Net (visual domain of real images)
  - Train on image classification
- Extend to Classification on a new category
  - – Q: How much can be reused?
- Fine-tune on new task
  - – E.g., semantic image segmentation
  - – > keep ‘backbone the same, fine-tune ‘head’ layers
  - – > assumption: visual features generalize within domain

Fine Tunining: Challenges

- Fine-tune on new task
  - Often need to train entire network, because input features will be different
  - Training only a few layers at the end is less likely to fundamentally solve it
- How much labelled data in the target domain?
  – Zero-shot learning
  – One-shot learning
  – Few-shot learning

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Why?

An example

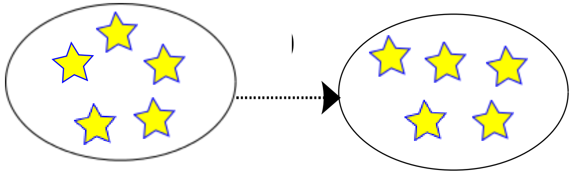
- We have **labeled** images from a **Web image** corpus
- Is there a `Person` in **unlabeled** images from a **Video** corpus ?

`Person`

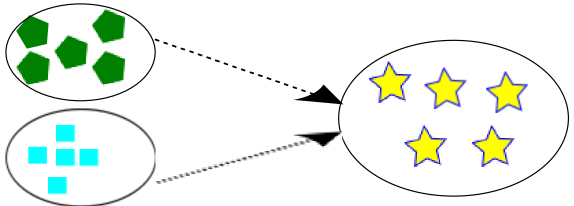
`no Person`

?

Is there a `Person`?



Training and test data are from the same domain



Training and test data are from different domains

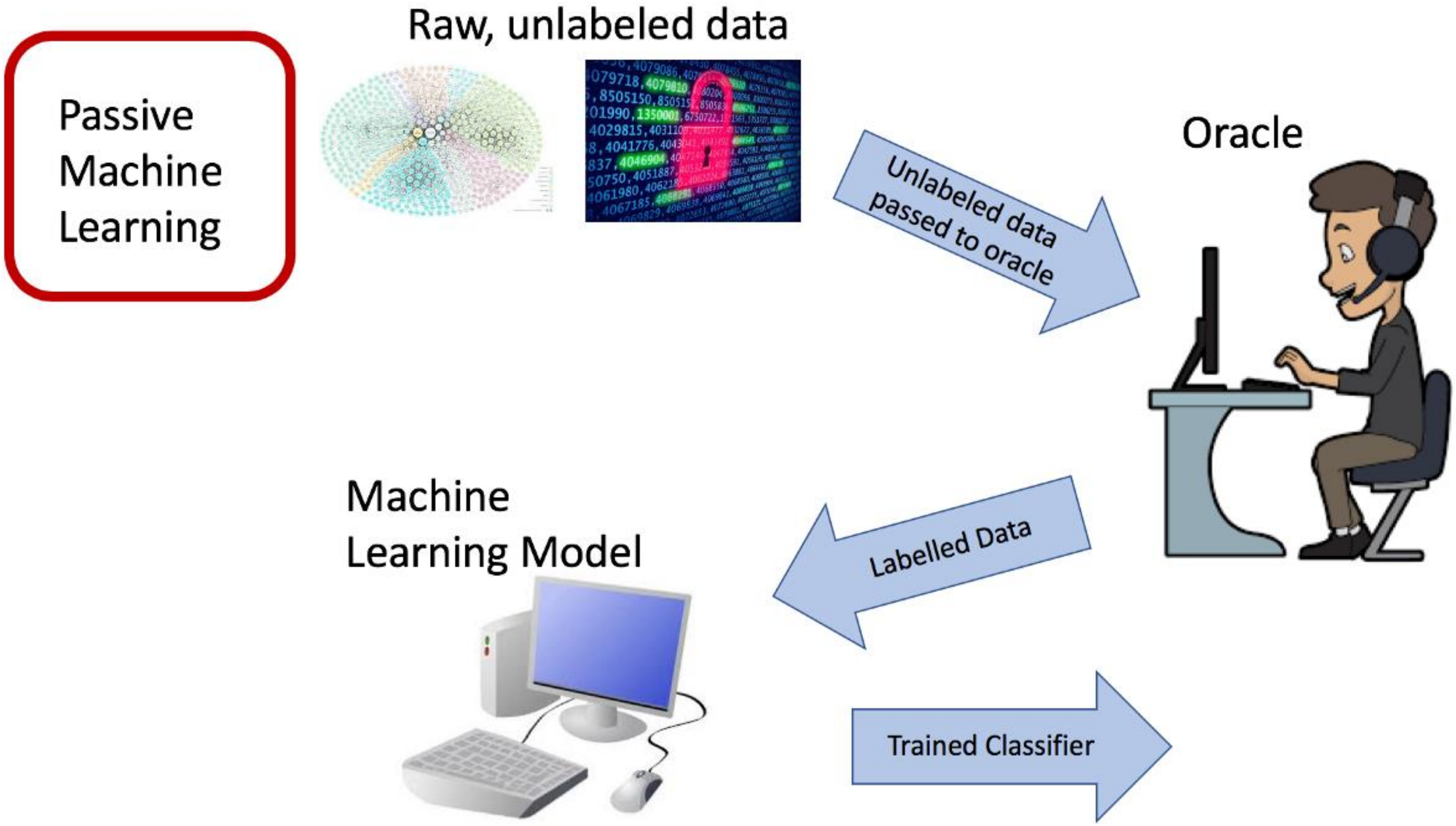
Challenge to Create in all situations

Proprietary

Private

Annotate Everything is "Impractical"

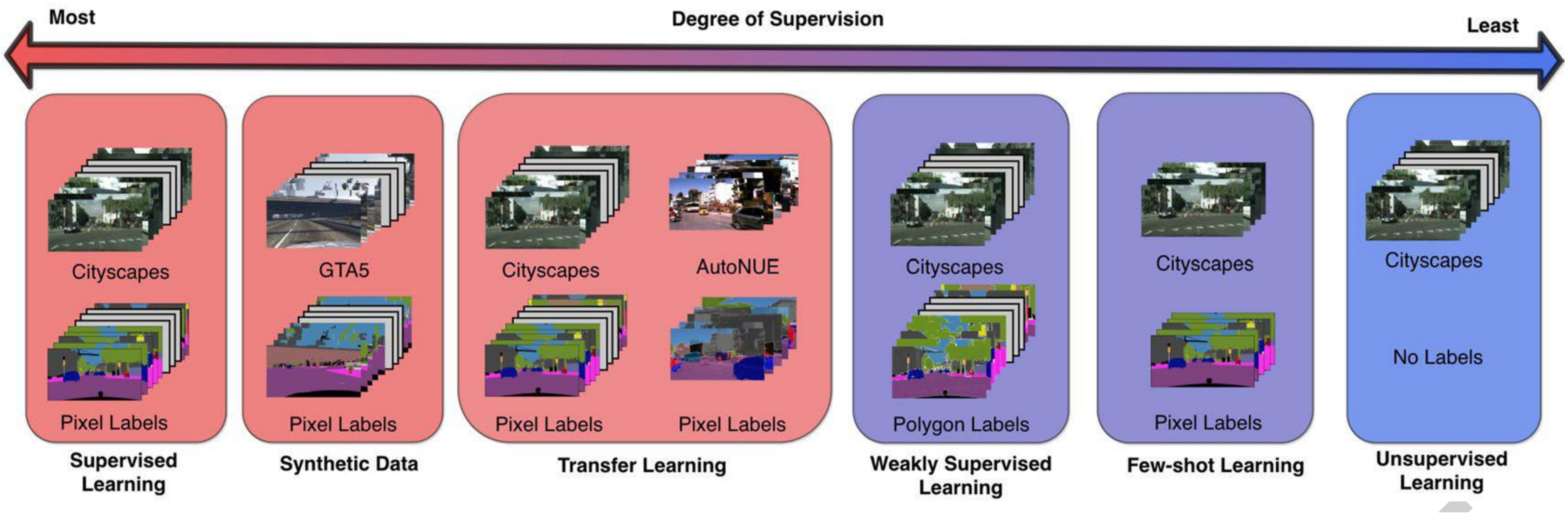
Active Learning



Example task: Semantic Segmentation

Expensive task and challenging to have millions of examples in all situations

Space of Supervision



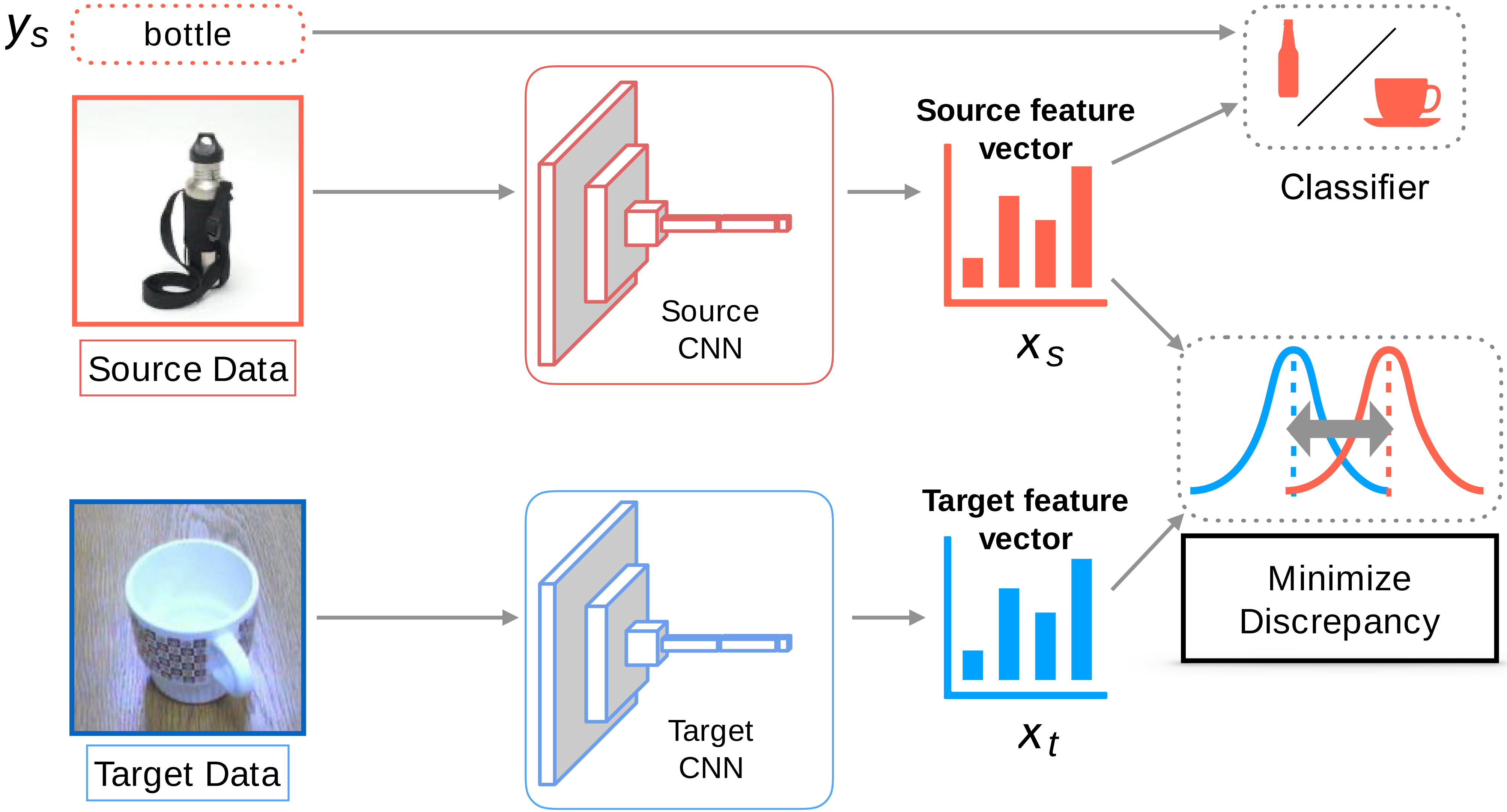
There is a need to adapt

Adapt

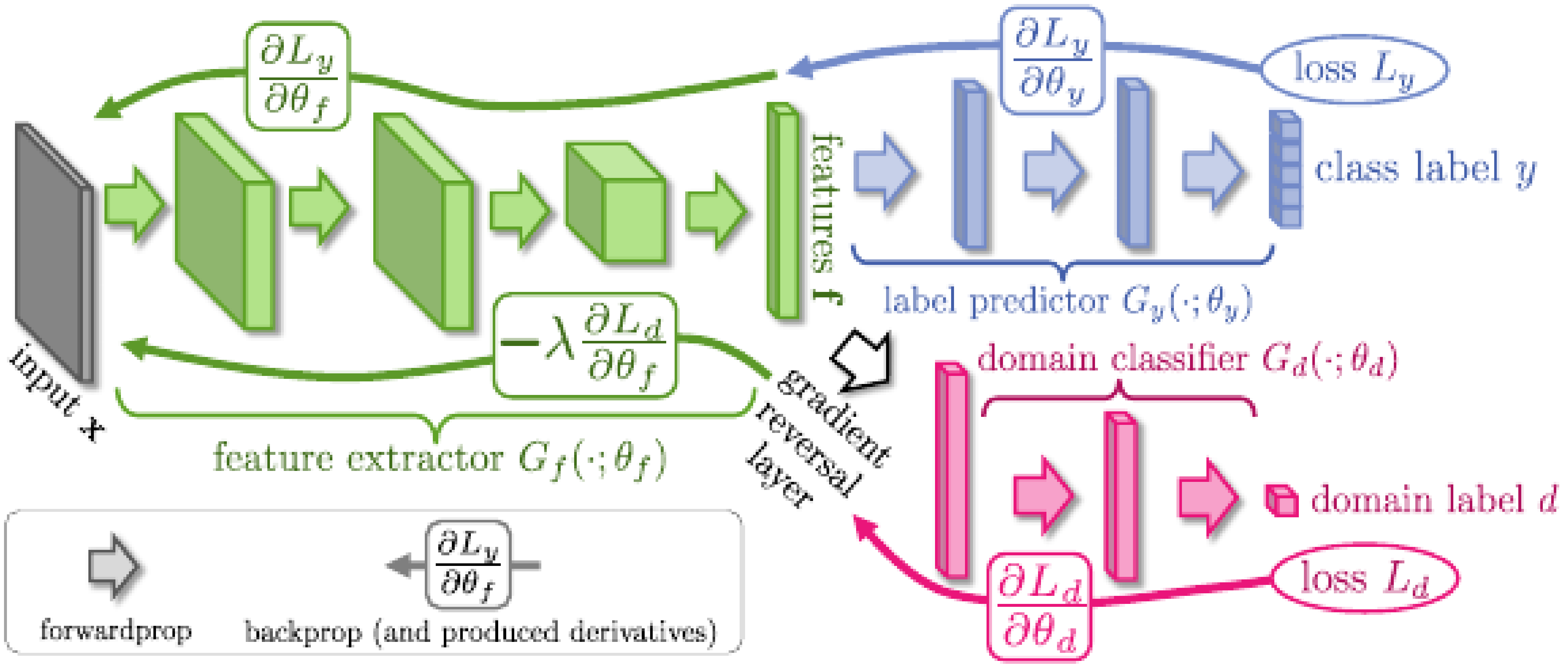
Source Domain: Lots of Training Data

Target Domain: Unlabeled or Limited Training Data

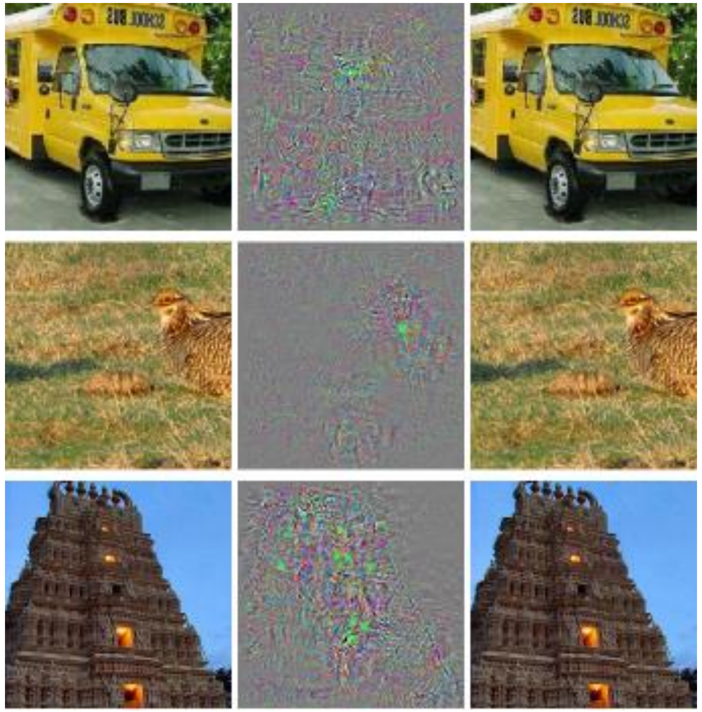
Deep Domain Adaptation

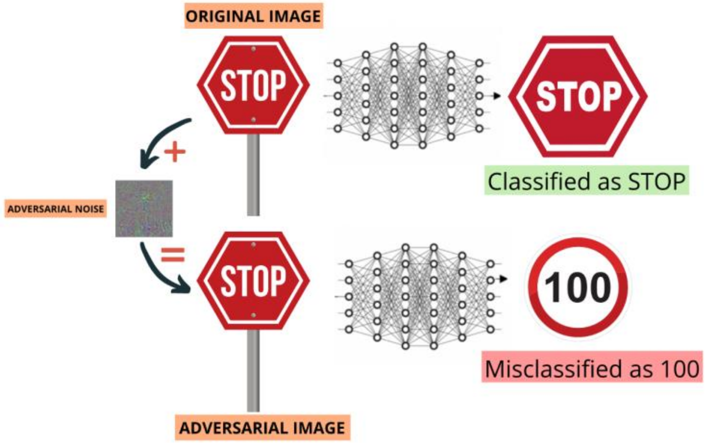


Domain Adversarial Adaptation



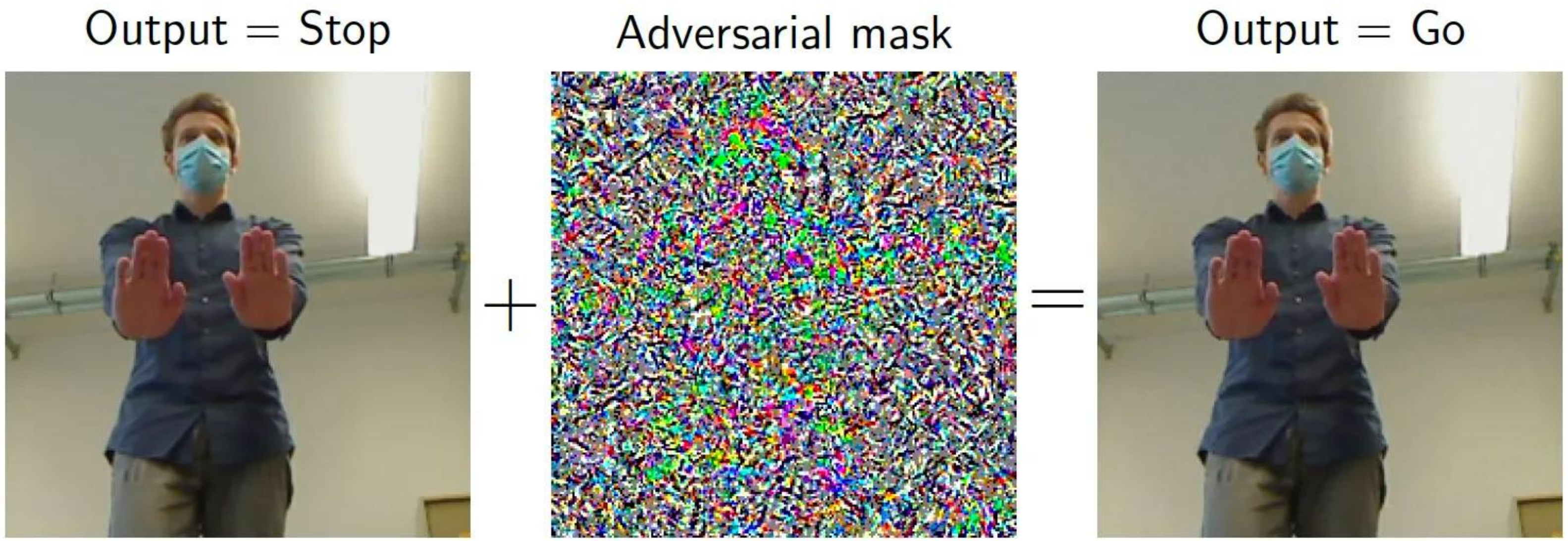
Neural Networks can be fooled?

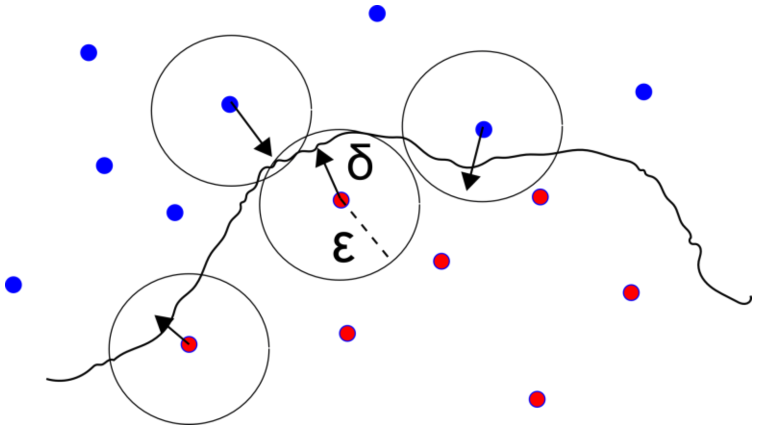




https://towardsdatascience.com/how-to-fool-a-neural-network-958ba5d82d8a

Role of Adversaries





Summary

- There are elegant and algorithmic results and methods available in many practical situation of our interest
  - Eg. Domain adaptation with no source data
- How do we approach in new situation?
  - Characterize the new situation properly
  - Learn to characterize the situation (than learn to solve the problem first)
- Often an effective strategy could be available already or directions for solving could be known well.

Thanks!!

Questions?